

# Multi-block statistics local kernel principal component analysis algorithm and its application in nonlinear process fault detection

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## ABSTRACT

It is vital for fault detection technology to extract features of industrial process data effectively. Local kernel principal component analysis (LKPCA) has proved its good performance in preserving global and local structural characteristics. However, it ignored useful high-order statistics of data, so multi-block statistics local kernel principal component analysis (MSLKPCA) algorithm integrating statistics pattern analysis (SPA) into LKPCA is proposed. The correlation coefficient matrix is first calculated and K-means clustering is adopted to divide the original variables into several blocks. Then the weighted SPA, which gives different weights to different samples in each window according to their distributions, is adopted to build statistic spaces containing both low-order and high-order statistics. After that, LKPCA is performed in each statistic space to realize feature extraction. To reduce the noise effect amplified by SPA, PCA is adopted in the residual space to remove noise. Bayesian strategy is used to fuse the results of each block and two monitoring statistics  $E_T$  and  $E_R$  are proposed to monitor the feature space and the residual space respectively. The Tennessee–Eastman (TE) process simulation shows the effectiveness and superiority of the proposed algorithm for process monitoring.

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## 1. Introduction

The development of science and technology has promoted the complexity and integration of modern industry, which leads to the restricted application of traditional multivariate statistical process monitoring techniques such as principal component analysis (PCA) [1], partial least squares (PLS) [2], fisher discriminant analysis (FDA) [3] and independent component analysis (ICA) [4] in industrial processes. For the purpose of enhancing the ability in fault detection, some manifold algorithms like locally linear embedding (LLE) [5], neighborhood preserving embedding (NPE) [6], Laplacian eigenmap (LE) [7], isometric feature mapping (ISPMAP) [8] and locality preserving projections (LPP) [9] have been proposed to preserve local nonlinear structure, while these algorithms cannot preserve global and local structures at the same time.

Recently, taking into account the global variance and local nonlinear characteristics, scholars have proposed some new linear dimensionality reduction algorithms. Zhang et al. proposed a global-local structure analysis (GLSA) method by establishing a new optimization objective function to obtain better fault detection results [10]. Yu proposed local and global principal component

analysis (LGPCA) with a different construction of objective function [11]. Tong et al. proposed an orthogonal multi-manifold projection (OMMP) algorithm that embeds neighbors' permutation information in sub-targets of the global structure and solves the problem of non-orthogonal feature vectors [12]. Shi et al. proposed a model based on ensemble structure analysis which combines PCA, LPP and MMP [13]. Luo et al. proposed a nonlocal and local structure preserving projection (NLSPP) algorithm, which can simultaneously preserve nonlocal and local structures with the use of nonlocal and local similarity weight coefficients [14]. Zhan et al. proposed global-local manifold analysis by incorporating statistical local approach into local and nonlocal embedding. These methods can achieve good results in linear systems, but unsatisfactory results in nonlinear processes. So some nonlinear algorithms have already been proposed [15]. Deng et al. proposed a local KPCA (LKPCA) algorithm that integrates local structure analysis with KPCA and constructs a new optimization objective which naturally involves both global and local structure information [16]. Cui et al. proposed ensemble LKPCA with different kernel width parameters [17]. Luo et al. proposed a kernel global-local preserving projections (KGLPP) algorithm, which performs more powerfully than KPCA and KLPP in capturing useful data characteristics [18]. She et al. proposed kernel orthogonal global-local preserving projections (KOGLPP) to overcome the non-orthogonality of KGLPP [19].

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With the increase in the dimension and complexity of industrial data, the high-order statistics are becoming more and more significant. He et al. proposed statistics pattern analysis (SPA) to compute low-order and high-order statistics to build statistic spaces of original data [20], which has already been applied in process monitoring. He et al. combined SPA and LPP to extract local and non-Gaussian features called statistics locality preserving projections (SLPP) [21]. Peng et al. combined SPA with kernel ICA (KICA) and introduced Markov distance to analyze the internal relations between samples, which is superior to the traditional feature decomposition monitoring algorithm [22]. Zhang et al. proposed local and global statistics pattern analysis (LGSPA) by integrating SPA into LGPCA [23]. Yang et al. proposed a method called balanced partial least square and SPA to promote the performance in process monitoring [24].

In order to extract features better and enhance the ability in fault detection of nonlinear process, a new algorithm called multi-block statistics local kernel principal component analysis is proposed. SPA is introduced to extract high-order statistics in the paper, which is beneficial to extract more features, but it also brings exorbitant dimensions since the dimension will multiply. In order to avoid the issue, original variables are divided into several blocks according to correlations between variables firstly. It not only reduces dimensions, but also facilitates the ability in feature extraction. Given that SPA amplifies the characteristics of original data accompanied by decaying ability in noise tolerance, weighted SPA is proposed to smooth the fluctuation caused by noise in the window based on samples' distribution and performed to construct different statistic spaces of each block secondly. Thirdly, LKPCA is adopted to project each statistic space into the feature space and the residual space. Since the fluctuation of data caused by noise are mostly distributed in the residual space, PCA is employed further to eliminate noise in the residual space. Then two monitoring statistics are constructed for monitoring. Eventually, Bayesian fusion strategy is used to integrate the performances of each block. The TE process validates that the proposed method has a better performance in nonlinear process fault detection.

The paper is arranged as follows. Section 2 introduces some algorithms including SPA and LKPCA. In Section 3, the variable partitioning strategy is mentioned first. Then MSLKPCA is proposed by introducing weighted SPA in LKPCA. Besides, two monitoring statistics for process monitoring are illustrated. Bayesian fusion strategy and process monitoring procedures are also introduced. The efficiency of the proposed method is demonstrated in Section 4. Finally, the work done in this paper and future work are summarized in Section 5.

## 2. Review of SPA and LKPCA

### 2.1. Statistics pattern analysis

High-order statistics of nonlinear process data usually contain some non-negligible characteristics, which cannot be extracted by traditional multivariate statistical method, so SPA has been proposed. Its procedure can be listed as follows [20]. Given a data set  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T \in \mathbf{R}^{n \times m}$ , where  $n$  is the number of samples and  $m$  is the number of variables. Utilize  $\mathbf{X}_c$  to denote a window of measurements:

$$\mathbf{X}_c = \begin{bmatrix} x_1(c-w+1) & x_2(c-w+1) & \cdots & x_m(c-w+1) \\ x_1(c-w+2) & x_2(c-w+2) & \cdots & x_m(c-w+2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(c) & x_2(c) & \cdots & x_m(c) \end{bmatrix} \quad (1)$$

where  $w$  and  $c$  represent the window width and the current sample index respectively. So  $\mathbf{X}_c$  can be seen as a subsampled set, and we can obtain a statistics pattern (SP) by calculating kinds of statistics of  $\mathbf{X}_c$ . The paper chooses four statistics to constitute an SP for less calculation and lower dimensionality, mentioned as follows.

Mean constitutes the first-order statistics:

$$\mu_i = \frac{1}{w} \sum_{l=0}^{w-1} x_i(c-l) \quad (2)$$

Variance constitutes the second-order statistics:

$$v_i = \frac{1}{w} \sum_{l=0}^{w-1} [x_i(c-l) - \mu_i]^2 \quad (3)$$

Other second-order statistics such as correlation, auto-correlation and cross-correlation are not considered because the paper focuses on the ability of processing non-linear data rather than describing the temporal relationships between samples.

Skewness and kurtosis constitute the high-order statistics:

$$\gamma_i = \frac{\frac{1}{w} \sum_{l=0}^{w-1} [x_i(c-l) - \mu_i]^3}{\left\{ \frac{1}{w} \sum_{l=0}^{w-1} [x_i(c-l) - \mu_i]^2 \right\}^{\frac{3}{2}}} \quad (4)$$

$$\kappa_i = \frac{\frac{1}{w} \sum_{l=0}^{w-1} [x_i(c-l) - \mu_i]^4}{\left\{ \frac{1}{w} \sum_{l=0}^{w-1} [x_i(c-l) - \mu_i]^2 \right\}^2} - 3 \quad (5)$$

where  $i = 1, 2, \dots, m$ . Use these statistics to construct an SP, i.e.  $\mathbf{x}_i^{\text{sp}} = [\mu_i, v_i, \gamma_i, \kappa_i]$ , and we can obtain series of SPs by sliding the window with a fixed step  $l_s$ , expressed as  $\mathbf{X}^{\text{sp}} = [\mathbf{x}_1^{\text{sp}}, \mathbf{x}_2^{\text{sp}}, \dots, \mathbf{x}_{n^{\text{sp}}}^{\text{sp}}]^T \in \mathbf{R}^{n^{\text{sp}} \times m^{\text{sp}}}$  with  $n^{\text{sp}}$  SPs and  $m^{\text{sp}}$  statistics.

### 2.2. Local kernel principal component analysis

Kernel principal component analysis (KPCA) is a nonlinear dimensionality reduction algorithm that introduces kernel function into PCA and performs better than PCA in nonlinear industrial process. Map low dimensional space to high dimensional feature space so that nonlinear space can be turned into linear space, which is beneficial to subsequent fault detection. Assuming a data set  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T \in \mathbf{R}^{n \times m}$ , a nonlinear map can be expressed as  $\mathbf{x}_i \in \mathbf{R}^{m \times 1} \rightarrow \Phi(\mathbf{x}_i) \in \mathbf{R}^{m \times 1}$ , where  $i = 1, 2, \dots, n$ . Then PCA can be performed in the feature space to find the projection vector  $\mathbf{p}$  that realizes maximizing variance. The projection of  $\mathbf{x}_i$  can be described as  $\mathbf{t} = \Phi(\mathbf{x}_i)^T \mathbf{p}$ , where  $\Phi(\mathbf{x}_i)$  is standardized with means and variances. Formulate the KPCA optimization task as [25]:

$$\begin{aligned} J_{\text{KPCA}}(\mathbf{p}) &= \max \frac{1}{n-1} \sum_{i=1}^n (\Phi(\mathbf{x}_i)^T \mathbf{p})^2 \\ \text{s.t. } \mathbf{p}^T \mathbf{p} &= 1 \end{aligned} \quad (6)$$

Form  $\mathbf{p}$  with series of coefficients  $\alpha_j$ ,  $1 \leq j \leq n$ :

$$\mathbf{p} = \sum_{j=1}^n \alpha_j \Phi(\mathbf{x}_j) \quad (7)$$

so Eq. (6) can be converted into:

$$\begin{aligned} J_{\text{KPCA}}(\boldsymbol{\alpha}) &= \max \frac{1}{n-1} \sum_{i=1}^n \left( \Phi(\mathbf{x}_i)^T \sum_{j=1}^n \alpha_j \Phi(\mathbf{x}_j) \right)^2 \\ &= \max \frac{1}{n-1} \sum_{i=1}^n \left( \sum_{j=1}^n \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) \right)^2 \end{aligned}$$

$$= \max \frac{1}{n-1} \alpha^T K K \alpha$$

$$\text{s.t. } \alpha^T K \alpha = 1 \quad (8)$$

where  $K$  is  $n \times n$  kernel matrix, which can be defined as  $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \Phi(\mathbf{x}_i)^T \Phi(\mathbf{x}_j)$ . Gauss kernel function  $k(\cdot)$  is chosen in the paper.

In order to analyze the inner relationship within data, local structure analysis (LSA) is introduced. Assuming the high-dimensional adjacent samples  $\Phi(\mathbf{x}_i)$  and  $\Phi(\mathbf{x}_j)$ , LSA aims at finding a project vector  $\mathbf{p}$  to keep low-dimensional samples  $\mathbf{y}_i = \Phi^T(\mathbf{x}_i)\mathbf{p}$  and  $\mathbf{y}_j = \Phi^T(\mathbf{x}_j)\mathbf{p}$  still being closed. Formulate the optimization task as [26]:

$$J_{\text{LSA}}(\mathbf{p}) = \min \sum_{i=1}^n \sum_{j=1}^n (y_i - y_j)^2 w_{ij}$$

$$= \min \sum_{i=1}^n \sum_{j=1}^n (\Phi(\mathbf{x}_i)^T \mathbf{p} - \Phi(\mathbf{x}_j)^T \mathbf{p})^2 w_{ij}$$

$$= \min \mathbf{p}^T \Phi^T(\mathbf{X})(\mathbf{D} - \mathbf{W})\Phi(\mathbf{X})\mathbf{p}$$

$$= \min \mathbf{p}^T \Phi^T(\mathbf{X})\mathbf{L}\Phi(\mathbf{X})\mathbf{p} \quad (9)$$

where  $\mathbf{L} = \mathbf{D} - \mathbf{W}$  is Laplacian matrix.  $\mathbf{D}$  is a diagonal matrix consisting of  $d_{ii} = \sum_{j=1}^n w_{ij}$ ,  $i = 1, 2, \dots, n$ .  $\mathbf{W}$  is an adjacent weighting matrix. Generally,  $\mathbf{W}$  can be decided by  $k$ -nearest neighbors (KNN) method [27], expressed as:

$$w_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_i, \mathbf{x}_j \in N_k(\mathbf{x}_i; \mathbf{x}_j) \\ 0, & \text{else} \end{cases} \quad (10)$$

where  $N_k(\mathbf{x}_i; \mathbf{x}_j)$  denotes the union of  $N(\mathbf{x}_i)$  and  $N(\mathbf{x}_j)$ .  $N(\mathbf{x}_i)$  and  $N(\mathbf{x}_j)$  represent the neighborhood of  $\mathbf{x}_i$  and  $\mathbf{x}_j$  respectively. Generally,  $N(\mathbf{x}_i)$  is calculated based on the Euclidean distance. Considering that LKPCA is adopted on SPs rather than original samples in the paper and adjacent SPs have stronger similarity because they share more samples,  $N(\mathbf{x}_i)$  is calculated based on the time-weighted Euclidean distance  $wd(\mathbf{x}_i, \mathbf{x}_j)$ , which is formulated as:

$$wd(\mathbf{x}_i, \mathbf{x}_j) = d(\mathbf{x}_i, \mathbf{x}_j) \times \exp\left(\frac{|i-j|}{t_d}\right) \quad (11)$$

where  $d(\mathbf{x}_i, \mathbf{x}_j)$  denotes the Euclidean distance between  $\mathbf{x}_i$  and  $\mathbf{x}_j$  and  $t_d$  is a positive constant used to prevent distant samples from overlarge weights. Samples closer in time may have a smaller weight leading to smaller distances.

Similar to KPCA,  $\mathbf{P}$  can be expressed with  $\alpha_j$ ,  $j = 1, 2, \dots, n$ , so Eq. (9) can be expressed as:

$$J_{\text{LSA}}(\alpha) = \min \alpha^T \Phi(\mathbf{X})\Phi(\mathbf{X})^T \mathbf{L}\Phi(\mathbf{X})\Phi(\mathbf{X})^T \alpha$$

$$= \min \alpha^T K L K \alpha \quad (12)$$

Introducing LSA into KPCA optimization and the LKPCA optimization can be formed as:

$$J_{\text{LKPCA}}(\alpha) = \max \frac{\alpha^T K K \alpha}{\alpha^T K L K \alpha}$$

$$\text{s.t. } \alpha^T K \alpha = 1 \quad (13)$$

The problem can be converted into solving the generalized Eigen-decomposition problem:

$$K K \alpha = \lambda K L K \alpha \quad (14)$$

a tiny positive regularization parameter can be added to the diagonal values of  $K L K$  when it is singular. Solve Eq. (14) and  $\alpha$  can be obtained.

### 3. Multi-block statistics local kernel principal component analysis algorithm for process monitoring

#### 3.1. Variable partitioning strategy

Pearson's correlation coefficient is an information-theoretic and is developed to measure the degree of correlation between two variables. Assuming two vectors  $\mathbf{x}_i$  and  $\mathbf{x}_j$ ,  $pcc(\mathbf{x}_i, \mathbf{x}_j)$  is formulated as [28]:

$$pcc(\mathbf{x}_i, \mathbf{x}_j) = \frac{\text{cov}(\mathbf{x}_i, \mathbf{x}_j)}{\sqrt{\text{var}(\mathbf{x}_i) \times \text{var}(\mathbf{x}_j)}} \quad (15)$$

where  $\text{cov}(\mathbf{x}_i, \mathbf{x}_j)$  is covariance and  $\text{var}(\cdot)$  is a variance. The  $pcc(\mathbf{x}_i, \mathbf{x}_j)$  ranges from  $-1$  to  $1$ , in which negative denotes negative correlations and positive denotes positive correlations. What the paper takes into consideration is the correlations, including negative and positive correlations. So the sign is abandoned and Pearson's correlation coefficient is calculated according to the following equation:

$$pcc(\mathbf{x}_i, \mathbf{x}_j) = \left| \frac{\text{cov}(\mathbf{x}_i, \mathbf{x}_j)}{\sqrt{\text{var}(\mathbf{x}_i) \times \text{var}(\mathbf{x}_j)}} \right| \quad (16)$$

A larger  $pcc(\mathbf{x}_i, \mathbf{x}_j)$  indicates that the two variables are more related.

Assuming original data  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T \in \mathbf{R}^{n \times m}$ , we can obtain the correlation matrix  $\mathbf{PCC}$ :

$$\mathbf{PCC} = \begin{bmatrix} pcc(\mathbf{x}_1, \mathbf{x}_1) & \cdots & pcc(\mathbf{x}_1, \mathbf{x}_m) \\ \vdots & \ddots & \vdots \\ pcc(\mathbf{x}_m, \mathbf{x}_1) & \cdots & pcc(\mathbf{x}_m, \mathbf{x}_m) \end{bmatrix} \quad (17)$$

the degree of correlation between variables can be observed from  $\mathbf{PCC}$ . K-means clustering is adopted to realize the partition of variables in the paper, and  $\mathbf{X}$  can be divided into  $b$  blocks  $\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots, \mathbf{X}^{(b)}$ .

#### 3.2. Multi-block statistics local kernel principal component analysis algorithm

SPA has advantages in magnifying data features. On the one hand, it is beneficial to detect faults, but on the other it leads to that an increasing sensitivity of the algorithm to noise. When a normal sample is disturbed by noise, it might be mistaken for a faulty sample. Hence two aspects are considered to decrease the situation occurs.

The first consideration is to smooth the impact caused by noise points on each window. Supposing the data distribution within a window is shown in Fig. 1, there are two noise points in the window. The lines in the Fig. 1 represent the distances between each sample point and its  $k_{spa}$  nearest neighbor. It is clear that the closer the distances, the denser the distributions of samples. If an SP is calculated on the basis of the original window, the two noise points may cause an alarm. Hence the weighted SPA using the weighted mean based on samples' distribution to substitute mean is proposed.

For  $\mathbf{X}_c$  in Eq. (1), calculate each samples'  $k_{spa}$  nearest distance in the window to constitute  $\mathbf{D} = [D_1, D_2, \dots, D_w]$ . Since the characteristics of a window depends on the most samples, at least half of the samples in each window are needed to judge whether each sample is one of the few noise points or not. So  $k_{spa}$  is chosen as half of  $w$ . The weighted value  $WD_q$  of each sample can be formed:

$$WD_q = \frac{1/D_q}{\sum_{q=1}^w 1/D_q} \quad (18)$$

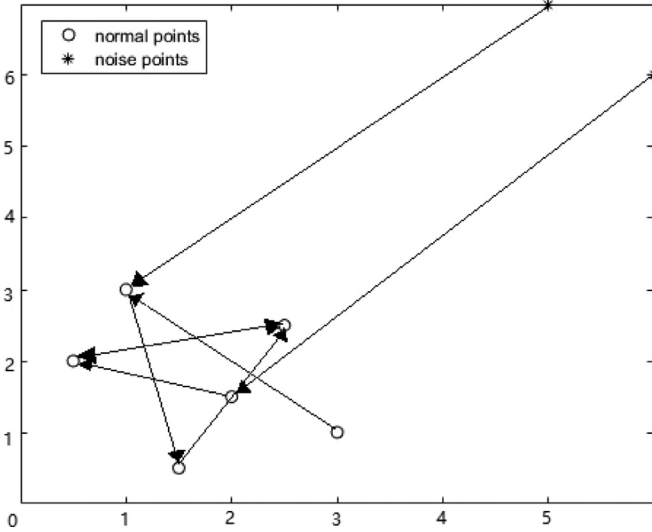


Fig. 1. Data distribution of a window.

where  $q = 1, 2, \dots, w$ . Small  $WD_i$  denotes a high possibility of being affected by noise, so it is necessary to reduce its weight when calculating the mean. The new mean in Eq. (1) is defined:

$$\mu'_i = \sum_{q=0}^{w-1} WD_{c-q} X_i(c-q) \quad (19)$$

where  $i = 1, 2, \dots, m$ . Meanwhile, the variance, skewness and kurtosis are all calculated based on the new mean.

Secondly, considering that data fluctuation caused by noise are mostly distributed in the residual space, a further step in residual space is taken to eliminate noise interference. Supposing  $b$  blocks  $\mathbf{X}^{(j)} \in \mathbf{R}^{n \times m^{(j)}}$ ,  $j = 1, 2, \dots, b$ ,  $b$  kernel matrixes  $\mathbf{K}^{SP(1)}, \mathbf{K}^{SP(2)}, \dots, \mathbf{K}^{SP(b)}$  of their responding statistic matrixes can be obtained. In addition to centralizing  $\mathbf{K}^{SP(j)}$ , the paper also takes variance scaling into consideration, which can be calculated as [29]:

$$\tilde{\mathbf{K}}^{SP(j)} = \mathbf{K}^{SP(j)} - \mathbf{K}^{SP(j)} \mathbf{I}_{n^{sp}} - \mathbf{I}_{n^{sp}} \mathbf{K}^{SP(j)} + \mathbf{I}_{n^{sp}} \mathbf{K}^{SP(j)} \mathbf{I}_{n^{sp}} \quad (20)$$

$$\tilde{\mathbf{K}}_S^{SP(j)} = \frac{\tilde{\mathbf{K}}^{SP(j)}}{\text{trace}(\tilde{\mathbf{K}}^{SP(j)}) / (n^{sp} - 1)} \quad (21)$$

where  $\mathbf{I}_n \in \mathbf{R}^{n \times n}$  and its elements are all  $1/n^{sp}$ .  $\tilde{\mathbf{K}}_S^{SP(j)}$  is obtained by scaling the centralized kernel matrix  $\tilde{\mathbf{K}}^{SP(j)}$ . The projection matrix  $\alpha_1^{(j)}, \alpha_2^{(j)}, \dots, \alpha_{n^{sp}}^{(j)}$  with  $\lambda_1, \lambda_2, \dots, \lambda_{n^{sp}}$  can be calculated by LKPCA algorithm. With series of  $\alpha_i^{(j)}$ ,  $\tilde{\mathbf{K}}_S^{SP(j)}$  can be decomposed into the feature space  $\mathbf{T}^{(j)}$  and the residual space  $\mathbf{E}^{(j)}$ :

$$\mathbf{T}^{(j)} = \sum_{i=1}^l \tilde{\mathbf{K}}_S^{SP(j)} \alpha_i^{(j)} \quad (22)$$

$$\mathbf{E}^{(j)} = \sum_{i=1+l}^{n^{sp}} \tilde{\mathbf{K}}_S^{SP(j)} \alpha_i^{(j)} \quad (23)$$

where  $l$  denotes the number of principal components of the  $j$ th block.

While the feature space extracts useful features including global and local information, the residual space retains most noise information. Conventionally, two monitoring statistics, i.e.  $T^2$  and  $Q$  statistics [30] can be built for fault detection. However, it is inappropriate to use  $Q$  statistic to monitor the residual space directly because of amplifying noise. Therefore PCA is performed further, then the transformation matrix  $\beta = [\beta_1^{(j)}, \beta_2^{(j)}, \dots, \beta_{n^{sp}-l}^{(j)}]$  can be

achieved. Sort  $\beta$  according to responding eigenvalues and select the first  $l_e$  vectors to realize noise filtering:

$$\mathbf{R}^{(j)} = \sum_{i_e=1}^{l_e} \mathbf{E}^{(j)} \beta_{i_e}^{(j)} = \sum_{i_e=1}^{l_e} \sum_{i=1+l}^{n^{sp}} \tilde{\mathbf{K}}_S^{SP(j)} \alpha_i^{(j)} \beta_{i_e}^{(j)} \quad (24)$$

where  $\mathbf{R}^{(j)}$  denotes the filtered residual space.

For a new sample  $\mathbf{x}_{new}$ , two monitoring indices are given depend on  $\mathbf{T}^{(j)}$  and  $\mathbf{R}^{(j)}$ :

$$T_t^{2(j)} = \mathbf{t}^{(j)T} (\mathbf{S}^{(j)})^{-1} \mathbf{t}^{(j)} \quad (25)$$

$$T_r^{2(j)} = \mathbf{r}^{(j)T} (\mathbf{S}_r^{(j)})^{-1} \mathbf{r}^{(j)} \quad (26)$$

where  $\mathbf{T}^{(j)} = [\mathbf{t}_1^{(j)}, \mathbf{t}_2^{(j)}, \dots, \mathbf{t}_l^{(j)}]$  and  $\mathbf{R}^{(j)} = [\mathbf{r}_1^{(j)}, \mathbf{r}_2^{(j)}, \dots, \mathbf{r}_{n^{sp}-l}^{(j)}]$ ,  $\mathbf{S}^{(j)} = \text{cov}(\mathbf{T}^{(j)})$  and  $\mathbf{S}_r^{(j)} = \text{cov}(\mathbf{R}^{(j)})$  is the covariance matrix of  $\mathbf{T}^{(j)}$  and  $\mathbf{R}^{(j)}$  respectively. Their confidence limits can be calculated by kernel density estimation (KDE) method [31]. Assuming a sample data set  $\mathbf{x} = [x_1, x_2, \dots, x_n]^T \in \mathbf{R}^n$ , a univariate kernel estimator is defined as:

$$\hat{f}(\mathbf{x}) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{\mathbf{x} - x_i}{h}\right) \quad (27)$$

where  $h$  is the kernel width. What's more,  $k(\cdot)$  denotes the chosen kernel function, which is often selected as Gaussian kernel function, so is the paper. Once the significant level is chosen, confidence limits  $\hat{T}_t^{2(j)}$  and  $\hat{T}_r^{2(j)}$  can be determined.

### 3.3. Bayesian strategy and ensemble learning

Since variables are divided into several blocks, it is necessary to combine their performance. Bayesian inference method can provide a pragmatic and intuitionistic solution [32], which is employed to combine different blocks' performance.

For the  $j$ th module, its statistics and corresponding control limits are  $T_t^{2(j)}$ ,  $T_r^{2(j)}$ ,  $\hat{T}_t^{2(j)}$  and  $\hat{T}_r^{2(j)}$  respectively. Bayesian inference is adopted to turn  $T_t^{2(j)}$  and  $T_r^{2(j)}$  into probabilities [17]:

$$p_t^{(j)}(\mathbf{F}|\mathbf{x}) = \frac{p_t^{(j)}(\mathbf{x}|\mathbf{F}) p_t^{(j)}(\mathbf{F})}{p_t^{(j)}(\mathbf{x})} \quad (28)$$

$$p_r^{(j)}(\mathbf{F}|\mathbf{x}) = \frac{p_r^{(j)}(\mathbf{x}|\mathbf{F}) p_r^{(j)}(\mathbf{F})}{p_r^{(j)}(\mathbf{x})} \quad (29)$$

where the probabilities  $p_t^{(j)}(\mathbf{x})$  and  $p_r^{(j)}(\mathbf{x})$  can be formulated as:

$$p_t^{(j)}(\mathbf{x}) = p_t^{(j)}(\mathbf{x}|\mathbf{N}) p_t^{(j)}(\mathbf{N}) + p_t^{(j)}(\mathbf{x}|\mathbf{F}) p_t^{(j)}(\mathbf{F}) \quad (30)$$

$$p_r^{(j)}(\mathbf{x}) = p_r^{(j)}(\mathbf{x}|\mathbf{N}) p_r^{(j)}(\mathbf{N}) + p_r^{(j)}(\mathbf{x}|\mathbf{F}) p_r^{(j)}(\mathbf{F}) \quad (31)$$

where N denotes normal condition and F denotes abnormal condition. Once the confidence level  $\alpha$  is determined,  $p_t^{(j)}(\mathbf{F})$ ,  $p_r^{(j)}(\mathbf{F})$ ,  $p_t^{(j)}(\mathbf{N})$  and  $p_r^{(j)}(\mathbf{N})$  can be identified as  $\alpha$ ,  $\alpha$ ,  $1-\alpha$  and  $1-\alpha$  respectively. The conditional probabilities  $p_t^{(j)}(\mathbf{x}|\mathbf{N})$ ,  $p_r^{(j)}(\mathbf{x}|\mathbf{F})$ ,  $p_t^{(j)}(\mathbf{x}|\mathbf{F})$  and  $p_r^{(j)}(\mathbf{x}|\mathbf{N})$  are defined as:

$$p_r^{(j)}(\mathbf{x}|\mathbf{N}) = \exp(-T_r^{2(j)} / \hat{T}_r^{2(j)}) \quad (32)$$

$$p_r^{(j)}(\mathbf{x}|\mathbf{F}) = \exp(-\hat{T}_r^{2(j)} / T_r^{2(j)}) \quad (33)$$

$$p_t^{(j)}(\mathbf{x}|\mathbf{N}) = \exp(-T_t^{2(j)} / \hat{T}_t^{2(j)}) \quad (34)$$

$$p_t^{(j)}(\mathbf{x}|\mathbf{F}) = \exp(-\hat{T}_t^{2(j)} / T_t^{2(j)}) \quad (35)$$

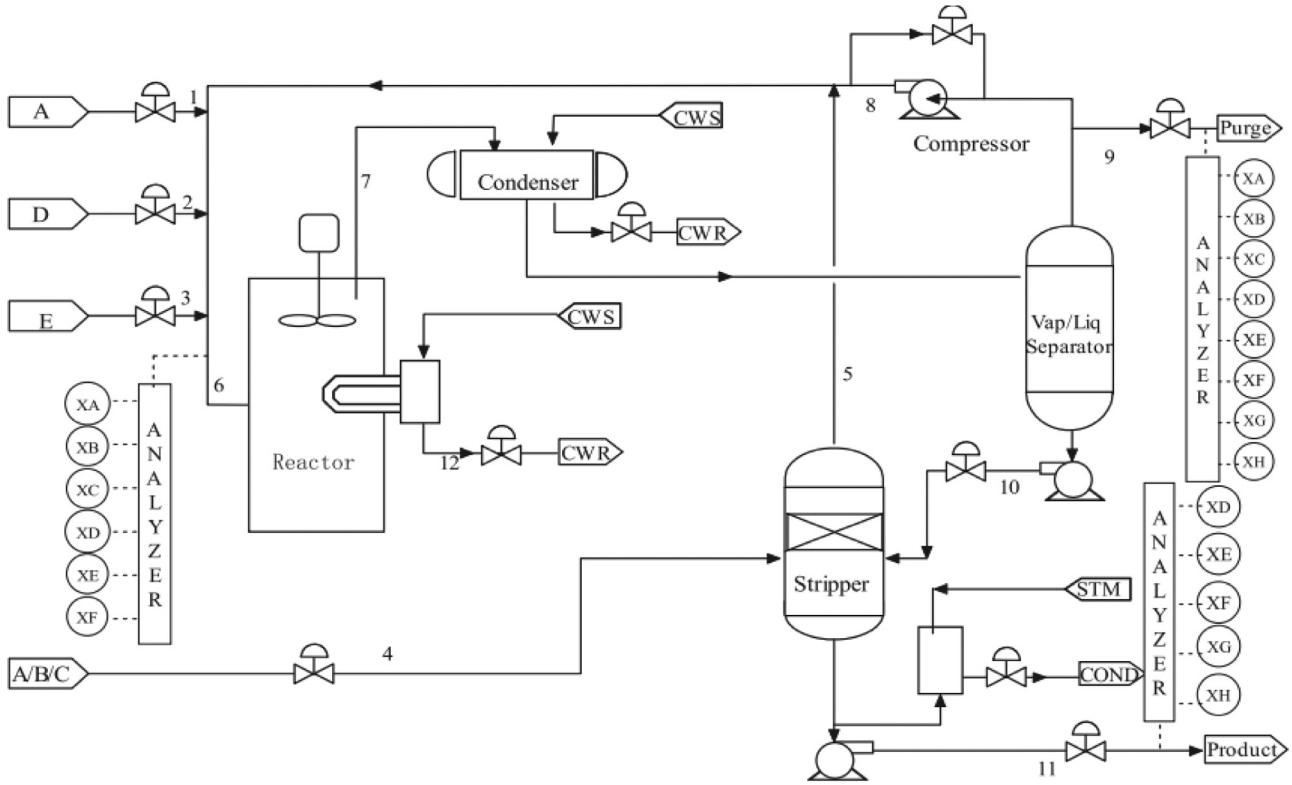


Fig. 2. Flowsheet of TE process.

Fusing the monitoring results of each sub-model and the final probabilities  $E_T$  and  $E_R$  can be obtained:

$$E_T = \sum_{i=1}^b \frac{p_t^{(i)}(\mathbf{F}|\mathbf{x})}{\sum_{j=1}^b p_t^{(j)}(\mathbf{F}|\mathbf{x})} p_t^{(i)}(\mathbf{F}|\mathbf{x}) \quad (36)$$

$$E_R = \sum_{i=1}^b \frac{p_r^{(i)}(\mathbf{F}|\mathbf{x})}{\sum_{j=1}^b p_r^{(j)}(\mathbf{F}|\mathbf{x})} p_r^{(i)}(\mathbf{F}|\mathbf{x}) \quad (37)$$

When  $E_T$  or  $E_R$  exceeds  $1-\alpha$ , abnormal condition is considered to have occurred.

### 3.4. Process monitoring procedure

The process monitoring procedure based on MSLKPCA consists of two steps: off-line modeling and on-line detection. Mainly including:

The off-line modeling:

- (1) Normalize training set  $\mathbf{X}$  and get  $\bar{\mathbf{X}}$ . Calculate Pearson's correlation coefficient within variables of  $\bar{\mathbf{X}}$  and then divide  $\mathbf{X}$  into  $b$  modules  $\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots, \mathbf{X}^{(b)}$  by K-means clustering;
- (2) Adopt weighted SPA to obtain statistic matrixes  $\mathbf{X}^{SP(j)} \in \mathbf{R}^{n^{sp} \times m^{sp(j)}}$ ,  $j = 1, 2, \dots, b$ . Normalize  $\mathbf{X}^{SP(j)}$  with its mean vector and covariance vector to get normalized matrixes  $\bar{\mathbf{X}}^{SP(j)}$ ;
- (3) Apply nonlinear mapping to obtain kernel matrixes  $\mathbf{K}^{SP(j)} \in \mathbf{R}^{n \times n}$ , centering and scaling  $\mathbf{K}^{SP(j)}$  to get  $\bar{\mathbf{K}}_S^{SP(j)}$ . Solve the generalized Eigen-decomposition problem to get projection matrixes;
- (4) Calculate  $T_t^{2(j)}$  and  $T_r^{2(j)}$  statistics of normal data, using KDE to calculate their corresponding confidence limits.

The on-line detection:

- (1) For the current sample  $\mathbf{x}_{new}$ , divide it into  $b$  blocks  $\mathbf{x}_{new}^{(1)}, \mathbf{x}_{new}^{(2)}, \dots, \mathbf{x}_{new}^{(b)}$ ;

- (2) Use the weighted SPA to calculate their statistic matrixes  $\mathbf{x}_{new}^{sp(j)}$ . When the number of collecting samples is less than  $w$ , training samples are used to make up for it. Normalize  $\mathbf{x}_{new}^{sp(j)}$  to get normalized matrixes  $\bar{\mathbf{x}}_{new}^{sp(j)}$ ;
- (3) Compute the kernel matrixes, centering and scaling it to get  $\bar{\mathbf{K}}_{new,s}^{sp(j)}$ . Then project them with responding projection matrixes and the statistics  $T_{t,new}^{2(j)}$  and  $T_{r,new}^{2(j)}$  can be obtained;
- (4) Combine each model's results with Bayesian strategy and get the final probabilities  $E_T$  and  $E_R$ . When they exceed  $1-\alpha$ , abnormal behavior is considered to have occurred. Otherwise, continue.

### 4. Case study

The TE process is provided to illustrate the validity and outperformance of the proposed method. Four methods including LKPCA, SKPCA and MSLKPCA are applied in process monitoring for comparison.

The TE process, composed of five units including reactor, condenser, separator, stripper and compressor, is a standard experimental simulation platform proposed by Downs and Vogel [33]. The detailed flow sheet can be seen in Fig. 2 Flowsheet of TE process. The process includes 12 operational variables and 41 manipulated variables (MVs), as shown in Table 1. 33 variables (1–22, 42–52) are monitored in this paper. A normal training set and 21 testing sets are used to test different algorithms, which can be downloaded from <https://github.com/camaramm/tennessee-eastman-profBraatz>. Details of the 21 faults are listed in Table 2. The training data set includes 500 normal samples while each testing data set includes 960 samples. Faults are introduced from the 161st sample.

Original variables are divided into  $b$  blocks according to the method mentioned in Section 3.1. Consider that each block has few variables when the parameter  $b$  is set too large, which is not only



**Table 1**  
list of the TE process variables.

| No    | Process variable                                 |
|-------|--------------------------------------------------|
| 1     | A feed (stream 1)                                |
| 2     | D feed (stream 2)                                |
| 3     | E feed (stream 3)                                |
| 4     | Total feed (stream 4)                            |
| 5     | Recycle flow (stream 8)                          |
| 6     | Reactor feed rate (Stream 6)                     |
| 7     | Reactor pressure                                 |
| 8     | Reactor level                                    |
| 9     | Reactor temperature                              |
| 10    | Purge rate (stream 9)                            |
| 11    | Product separator temperature                    |
| 12    | Product separator level                          |
| 13    | Product separator pressure                       |
| 14    | Product separator underflow (stream 10)          |
| 15    | Stripper level                                   |
| 16    | Stripper pressure                                |
| 17    | Stripper underflow (stream 11)                   |
| 18    | Stripper temperature                             |
| 19    | Stripper steam flow                              |
| 20    | Compressor work                                  |
| 21    | Reactor cooling water outlet temperature         |
| 22    | Separator cooling water outlet temperature       |
| 23–28 | Components A, B, C, D, E and F in stream 6       |
| 29–36 | Components A, B, C, D, E, F, G and H in stream 9 |
| 37–41 | Components D, E, F, G and H in stream 11         |
| 42    | MV for D feed flow (stream 2)                    |
| 43    | MV for E feed flow (stream 3)                    |
| 44    | MV for A feed flow (stream 1)                    |
| 45    | MV for total feed flow (stream 4)                |
| 46    | MV for compressor recycle valve                  |
| 47    | MV for purge valve (stream 9)                    |
| 48    | MV for separator pot liquid flow (stream 10)     |
| 49    | MV for stripper liquid prod flow (stream 11)     |
| 50    | MV for stripper steam valve                      |
| 51    | MV for reactor cooling water flow                |
| 52    | MV for condenser cooling water flow              |

**Table 2**  
TE process fault patterns.

| Fault ID | Process variable                                      | Fault type       |
|----------|-------------------------------------------------------|------------------|
| 1        | A/C feed ratio, B composition constant (stream 4)     | Step             |
| 2        | B composition, A/C feed ratio constant (stream 4)     | Step             |
| 3        | D feed temperature (stream 2)                         | Step             |
| 4        | Reactor cooling water inlet temperature               | Step             |
| 5        | Condenser cooling water inlet temperature             | Step             |
| 6        | A feed loss (stream 1)                                | Step             |
| 7        | C header pressure loss-reduced availability(stream 4) | Step             |
| 8        | A, B and C feed compositions (stream 4)               | Random variation |
| 9        | D feed temperature (stream 2)                         | Random variation |
| 10       | C feed temperature (stream 4)                         | Random variation |
| 11       | Reactor cooling water inlet temperature               | Random variation |
| 12       | Condenser cooling water inlet temperature             | Random variation |
| 13       | Reaction kinetics                                     | Slow shift       |
| 14       | Reactor cooling water valve                           | Sticking         |
| 15       | Condenser cooling water valve                         | Sticking         |
| 16–20    | Unknown                                               | Unknown          |
| 21       | Valve position constant (stream 4)                    | Constant         |

increase the influence of variable fluctuations on the performance, but also increase the calculation amount. So the paper divides 33 variables into 4 blocks, as shown in Table 3.

Three methods including LKPCA, SPCA and MSLKPCA are used for fault detection. The Gaussian kernel function parameters in LKPCA and MSLKPCA can be set according to the number of vari-

**Table 3**  
Partition results of the original variables.

| Variable block | Variable label                               |
|----------------|----------------------------------------------|
| 1              | 2,9,22,23,32                                 |
| 2              | 7,13,16,20,27                                |
| 3              | 10,11,18,19,28,31                            |
| 4              | 1,3,4,5,6,8,12,14,15,17,21,24,25,26,29,30,33 |

**Table 4**  
MAR (%) of three methods for 21 faults.

| Fault ID | LKPCA |       | SPCA  |      | MSLKPCA |       |
|----------|-------|-------|-------|------|---------|-------|
|          | $T^2$ | $Q$   | $T^2$ | $Q$  | $E_T$   | $E_R$ |
| 1        | 0.13  | 0.13  | 0.13  | 0    | 0.25    | 0.25  |
| 2        | 1.25  | 1.5   | 1.63  | 0    | 1.25    | 2     |
| 3        | 91.5  | 95.5  | 94.38 | 0.5  | 78.88   | 81.38 |
| 4        | 2     | 0     | 13.38 | 0    | 0       | 0     |
| 5        | 68    | 0     | 71.63 | 0    | 0       | 0     |
| 6        | 0     | 0     | 0     | 0    | 0       | 0     |
| 7        | 0     | 0     | 0     | 0    | 0       | 0     |
| 8        | 1.5   | 2     | 2.25  | 0    | 1.13    | 1.5   |
| 9        | 91.63 | 96.25 | 96    | 0.13 | 82.25   | 85.25 |
| 10       | 23.25 | 16.38 | 37.75 | 0    | 8.63    | 8.38  |
| 11       | 23.75 | 25.88 | 6     | 0    | 1.25    | 1.63  |
| 12       | 0.88  | 0.25  | 0.25  | 0    | 0.25    | 0.25  |
| 13       | 4.75  | 4.38  | 5.13  | 0    | 4.25    | 4.75  |
| 14       | 0     | 0     | 0.13  | 0    | 0.13    | 0.13  |
| 15       | 84    | 93.63 | 94.75 | 0.63 | 78.13   | 82.63 |
| 16       | 23    | 10.75 | 33.13 | 0    | 3       | 4.13  |
| 17       | 10.25 | 2.75  | 4.63  | 0    | 2.13    | 2.75  |
| 18       | 10.13 | 9.38  | 11    | 0    | 9.25    | 9.88  |
| 19       | 56.88 | 10.5  | 3.75  | 0    | 0.38    | 3.63  |
| 20       | 29.38 | 24.38 | 36.63 | 0.13 | 10.38   | 20.25 |
| 21       | 44.38 | 50.75 | 64.25 | 0    | 60.13   | 44.88 |
| AMAR     | 26.98 | 21.16 | 27.46 | 0.07 | 16.27   | 16.84 |

ables [33]. Besides, the paper selects  $l_s$  as 1 for real-time performance and dynamic property. The parameter  $w$  is fixed as 10 to study the effects of different  $k$  in each block on the detection results. Select the most sensitive faults of each module, i.e. fault 4, 11 and 14 for block 1, fault 8, 13 and 20 for block 2, fault 1, 5, 12 and 21 for block 3 and fault 1, 10 and 18 for block 4. Calculate the two commonly used indices, false alarming rate (FAR) and average missing alarming rate (MAR) of two statistics under different  $k$  values, as shown in Fig. 3. Obviously, the increase of  $k$  value is accompanied by the increase of FAR. So the paper sets  $k$  as 4, 4, 3 and 3 for block 1–4 in MSLKPCA respectively. The  $k$  value in LKPCA is set according to literature [16] for better results. In addition, there is also no conclusion about how to set  $w$ , which should take the effect of the fault on the process and its magnitude as consideration. In actual, different parameters should be set for different faults, but it is not realistic because we don't know what kind of fault is going to occur, and we can't adjust  $w$  according to the forthcoming fault. Hence the paper set the parameter according to the average performance of MSLKPCA under different  $w$  and selects fault 1, 5, 8, 10, 11, 14, 19 and 21 that are sensitive to  $w$  as representatives to determine  $w$  value. In order to prevent high FAR caused by overlarge  $w$ , this paper chooses  $w$  ranging from 1 to 50. Use the average MAR and the average FAR of  $E_T$  and  $E_R$  statistics for those faults to represent different performances under different  $w$ , and lower MAR and lower FAR mean better performance. The trend of average FAR and MAR with respect to  $w$  is shown in Fig. 4. FAR presents an overall growing trend while MAR presents a fluctuating trend accompanied by increasing  $w$ , which means FAR is more sensitive to  $w$  than MAR. So the paper sets  $w$  as 12 with integrated consideration of stability, low FAR and low MAR, and the same  $w$  for SPA for comparison. The number of PCs is determined according to cumulative percent value (CPV) greater than

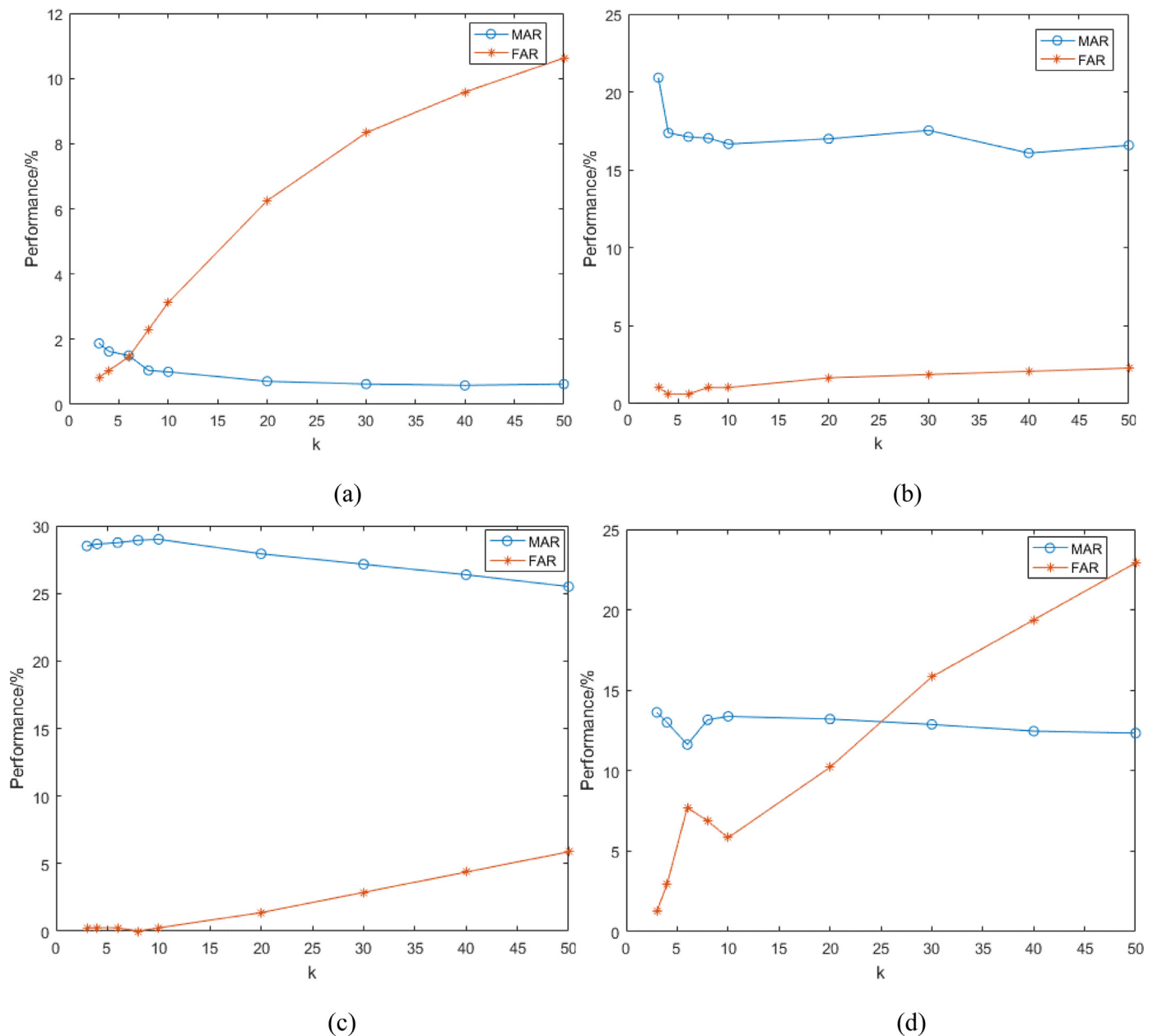


Fig. 3. The performance of (a) block 1, (b) block 2, (c) block 3 and (d) block 4 with different  $k$ .

Table 5

Average FAR (%) of three methods for 21 faults.

|      | LKPCA |      | SPCA  |       | MSLKPCA |       |
|------|-------|------|-------|-------|---------|-------|
|      | $T^2$ | $Q$  | $T^2$ | $Q$   | $E_T$   | $E_R$ |
| AFAR | 2.35  | 3.27 | 2.08  | 95.74 | 5.51    | 4.43  |

85% for all methods. The confidence limit is set to 99% for monitoring statistics.

Table 4 shows MARs of 21 faults of three methods. It should be noticed that due to different training sets and parameters, some detection results of LKPCA are different from those in literature [16]. In general, three methods can detect fault 1, 2, 6, 7, 8, 12, 13 and 14 better while have poor performance on fault 3, 9 and 15 since the three faults have little impact on the process variables [34]. Take a holistic look of monitoring results, MSLKPCA has a better performance than the other methods, whose average

MAR (AMAR) of  $E_T$  and  $E_R$  statistics are 16.27% and 16.84%. The  $T^2$  and  $Q$  statistics of LKPCA and SPCA are 26.98%, 21.16% and 27.46%, 0.07%, which validates that the proposed algorithm is more sensitive to faults. Since both lower-order and high-order statistics, local structure and global structure are taken into account simultaneously, the ability in extracting useful information is strengthened. Besides, the weighted SPA and further step in the residual space bring the advantage of SPA and avoid its disadvantage. We can observe that though the  $Q$  statistic of SPCA has low AMAR, its average FAR (AFAR) is 95.74% according to Table 5, which is consistent with what was mentioned in Section 3.2. On the contrary, the AFAR of the  $E_R$  statistic is just 4.43%, explaining the effectiveness and necessity of adopting PCA for noise reduction in the residual space. So the  $Q$  statistic of SPCA will not be evaluated in the paper because of its bad description of normal process. Specifically, compared with LKPCA and SPCA, MSLKPCA has an obvious advantage in fault 5, 10, 11, 16, 19 and 20. Fault 5, 10 and 19 are used for a concrete analysis.

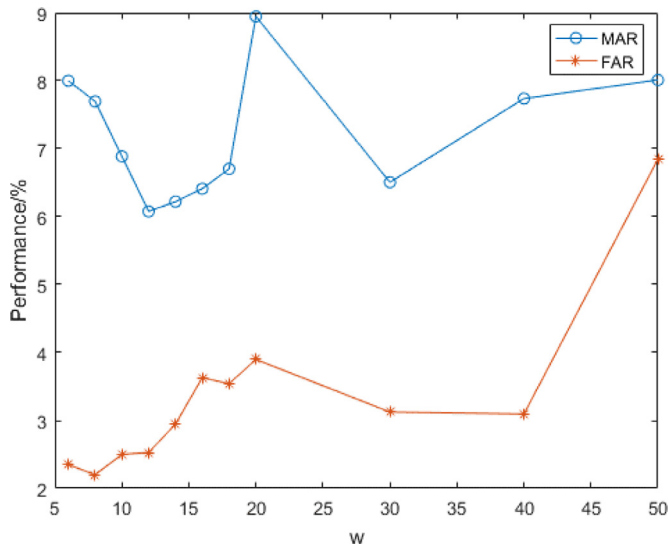


Fig. 4. Different performances of MSLKPCA under different  $w$ .

Fault 5 is a step change in condenser cooling water inlet temperature. Three monitoring plots are shown in Fig. 5. Since there is a feedback loop to control the effect of the fault on process variables, the  $T^2$  statistic of LKPCA and SPCA return below corresponding control limits from about the 367th sample, meaning that many fault samples are failed to be detected. In contrast, both  $E_T$  and  $E_R$  statistics can detect the variation timely without miss-

ing faulty samples and greater than the responding control limits continuously.

Fault 10 describes a random change in the temperature of material c. As fault 10 is a random fault, it is difficult for LKPCA to detect the fault when the changes of a single sample's variables are not obvious. Nevertheless, MSLKPCA analyzes variations of all samples in the window rather than a single sample, which is favorable for obtaining more seasonable and comprehensive and information. Monitoring plots of three methods are shown in Fig. 6, from which we can see that many faulty samples have not been detected by SPCA and LKPCA. The MARs of their  $T^2$  statistics are more than 15%. However, the  $E_T$  and  $E_R$  statistics of MSLKPCA can detect the fault at 186th and 188th sample respectively with MAR equaling to about 8%, showing better performance.

Fault 19 is a fault with an unknown type. The monitoring results are listed in Fig. 7. As shown in the figure, the  $Q$  statistic of LKPCA shows better performance than  $T^2$  statistic, with MAR equaling to 10.5%. The  $T^2$  statistic shows a smaller MAR, i.e. 3.75%. However, their detected performance are not as good as MSLKPCA. The MAR of the  $E_T$  statistic is just 0.38% and stays above the control limit after the fault being detected. Compared with LKPCA and SPCA, MSLKPCA shows a satisfying result and the superiority in fault 19.

In fact, practical nonlinear processes are often disturbed by noise, especial measurement noises in multi-sensor environment [35,36]. The paper takes fault 10 as an example to study the feasibility of MSLKPCA under the disturbance of Gaussian noise considering that the influence of noise on random faults may be greater than that on faults with obvious variations. Gaussian

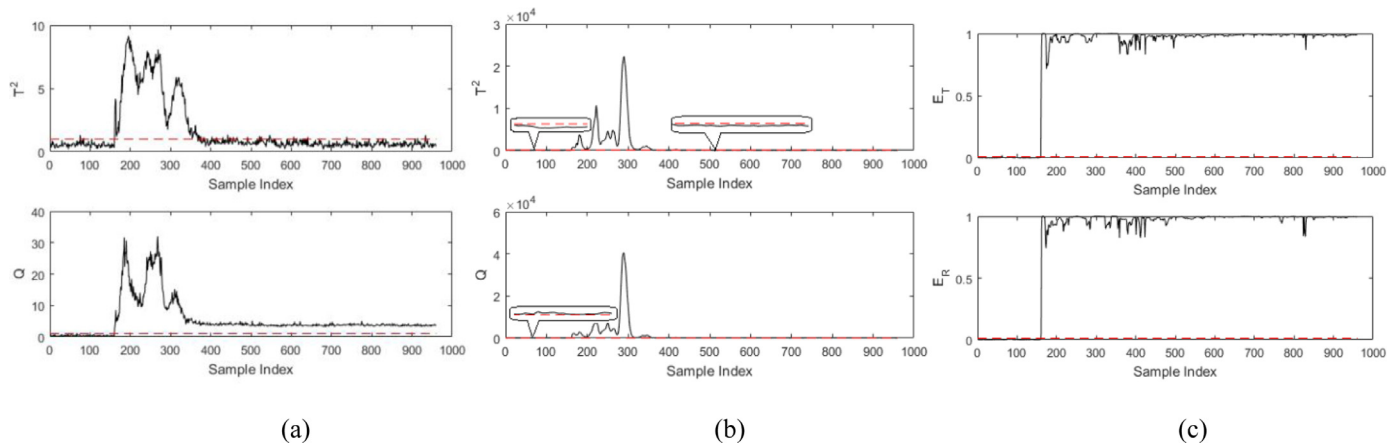


Fig. 5. Monitoring results for fault 5 by (a) LKPCA (b) SPCA and (c) MSLKPCA.

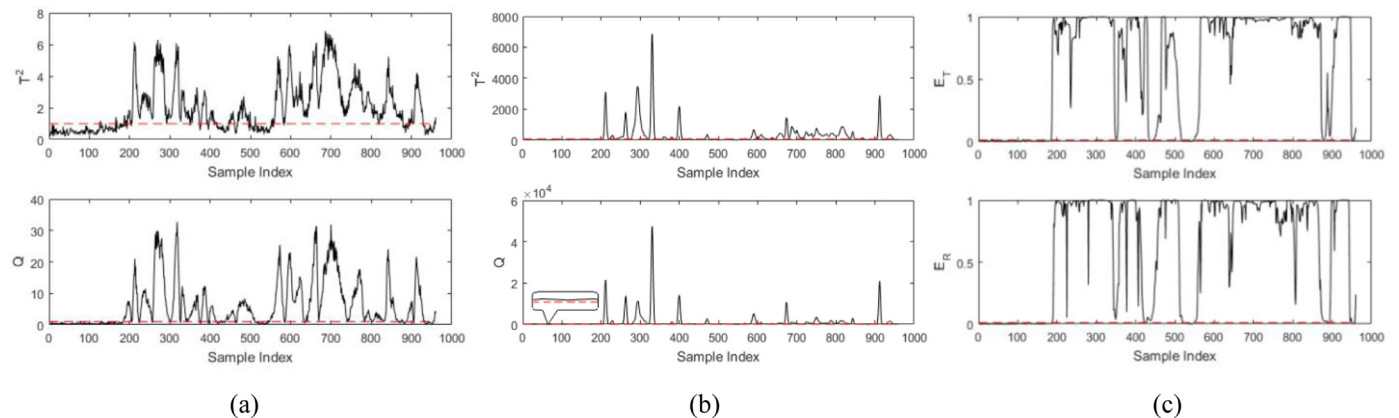


Fig. 6. Monitoring results for fault 10 by (a) LKPCA (b) SPCA (c) MSLKPCA.



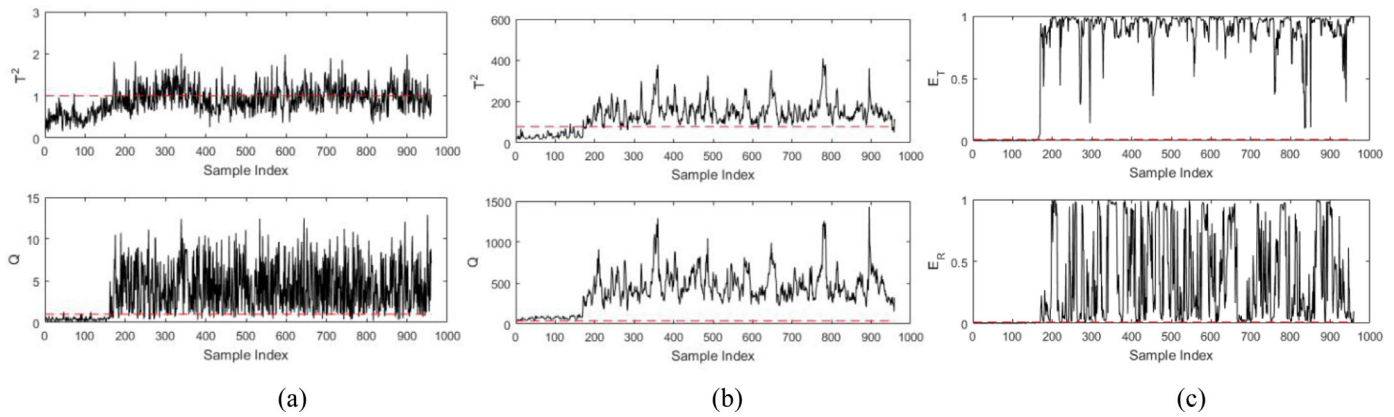


Fig. 7. Monitoring results for fault 19 by (a) LKPCA (b) SPCA and (c) MSLKPCA.

**Table 6**  
MAR (%) of MSLKPCA under different Gaussian noises.

| Variance | $E_T$ | $E_R$ |
|----------|-------|-------|
| 0.005    | 7.63  | 9.88  |
| 0.01     | 8.13  | 10.88 |
| 0.02     | 14    | 11.13 |
| 0.05     | 16.5  | 18.88 |
| 0.1      | 16.13 | 21.63 |

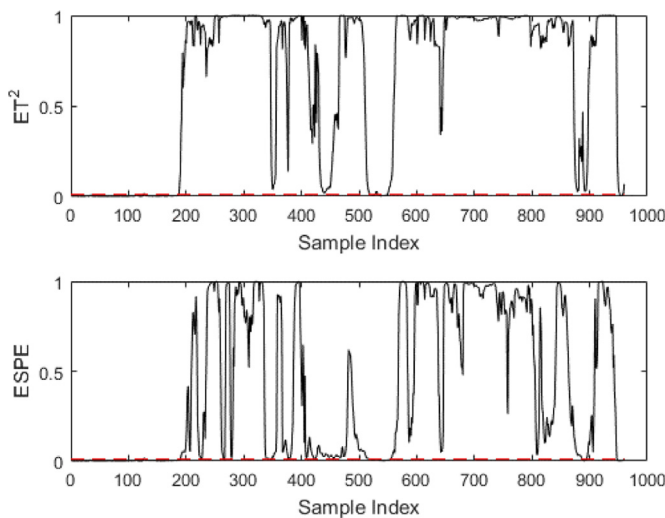


Fig. 8. Monitoring results for fault 10 with Gaussian noise.

measurement noises with mean equaling to 0 and variance ranging from 0.005 to 0.1 are added to these measured variables with regard to temperature, which are variable 9, 11 and 18 for test. The results are shown in Table 6, from which we can observe that MARs of both statistics are not much different from the results in Table 4 when the noise variance is within 0.01. Fig. 8 shows that the monitoring performance of Gaussian noises with variance being 0.01. It can be seen that only small changes occurred in the  $E_R$  statistic by comparing Fig. 8 with Fig. 6(c). While variances are greater than 0.01, the MARs of both statistics start to increase. So the proposed algorithm shows some robustness to Gaussian noises within the acceptable range.

## 5. Conclusion

The paper proposes a new algorithm named as multi-block statistics local kernel principal component analysis for nonlinear

process monitoring. Not only does MSLKPCA make full use of low-order and high-order statistics of process data for better feature extraction, but it also preserves local structure and global structure. Original variables are firstly divided into several blocks based on the collection within variables and K-means clustering. Since SPA enhances fault detection performance by magnifying features, false alarming samples may increase at the same time because of the fluctuation of normal samples caused by noise. So weighted SPA and further processing using PCA to eliminate noise in the residual space are proposed, which is beneficial to reduce the impact of noise and keep low MAR and low FAR simultaneously. The monitoring results of the TE process based on LKPCA, SPCA and MSLKPCA show that the proposed method outperforms the other methods.

It should be mentioned that the proposed algorithm is suitable for single mode while multiple modes are more prevalent in industrial processes because industries need to cater for the needs of market. So how to enhance the robustness of the proposed algorithm to noise and apply it in multi-mode process and distinguish different modes will be the focus in future studies.

## Declaration of Competing Interest

None.

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